PREDICTIVE ANALYTICS WITH ATHLETICS DATA

Week 3 · Beyond statistical significance: Bayes' theorem

TUESDAY SEP 8 AND THURSDAY SEP 10, 2026

OBJECTIVES

WHAT YOU WILL BE ABLE TO DO BY MONDAY NIGHT

1. Differentiate between “how surprising the data is if nothing changed” and “the likelihood that something did change”.
2. Identify conditions where prior beliefs should be included in analysis. Then, quantify those prior beliefs and factor them in using Bayes' theorem.

UPDATE BOTH FOLDERS BEFORE WE START

1. Open Claude Code in your repo folder, spax402-your-username.

   Your spax402 folder holds two folders. This one is your personal repository, where you commit your work. Next to it is spax402-course-materials, our shared materials, which Professor Davis updates weekly.

2. Run one prompt.

   The /update skill brings this week's files into your personal repo without saving over your work, and the course materials folder updates by asking Claude to pull the latest version.

   > Run /update to refresh my repo, then pull the latest course materials in the folder next to mine, spax402-course-materials. ↵

3. Confirm that weeks/week03/ now exists in your repo.

   It holds this week's README, the worksheet you fill in on Thursday, and the Case Study data.

01

A QUARTER ON THE SIDEWALK

You find a quarter on the sidewalk, flip it five times, and get five heads.

? IS THERE SIGNIFICANT EVIDENCE THAT THE COIN IS WEIGHTED?

For simplicity, assume a weighted coin turns up heads every time.

? DO YOU THINK THERE WAS SIGNIFICANT EVIDENCE THAT THE COIN IS WEIGHTED? WHY OR WHY NOT?

WHAT IS 3.125%?

It is not the probability the coin is fair.

It assumes the coin is fair and gives the probability of the result.

The frequentist approach sets a binary threshold of 0.05 for statistical significance.

Above it, nothing happened. Below it, something did.

What does this ignore?

BAYESIAN DEFINITIONS

	WHAT IT MEANS	THE SIDEWALK QUARTER
Prior	What you believed before you saw the data	My best guess is that 1 in 100,000 coins on the ground is weighted, so a 99.999% chance the coin is fair
Likelihood	How probable the data is, under each hypothesis	3.125% if fair, 100% if weighted
Posterior	What you believe after you have seen the data	?

$$\underbrace{P(H \mid \text{data})}_{\text{posterior}} = \frac{\underbrace{P(H)}_{\text{prior}} \times \underbrace{P(\text{data} \mid H)}_{\text{likelihood}}}{P(\text{data})}$$

? SAME COIN, SAME FIVE HEADS, BUT YOU BOUGHT IT IN A MAGIC SHOP. WHAT CHANGES?

THE MORAL

	STATISTICAL SIGNIFICANCE	POSTERIOR PROBABILITY
Asks	If the null hypothesis were true, how surprising is my data?	Given my data, how likely is this hypothesis to be true?
Prior beliefs	Ignored	Incorporated
Answer	A p-value, judged against 0.05	A probability of the hypothesis

Evidence should be considered, but our prior belief should impact how we interpret evidence.

TO CONTINUE BUILDING INTUITION AROUND BAYES' THEOREM, WATCH THIS VIDEO.

Bayes theorem, the geometry of changing beliefs, 3Blue1Brown.

https://www.youtube.com/watch?v=HZGCoVF3YvM

02

A SLUMP OR A CLIFF?

SAQUON BARKLEY, YARDS PER CARRY BY SEASON

SEASON	TEAM	CARRIES	YARDS PER CARRY
2018	NYG	263	4.98
2019	NYG	217	4.62
2020	NYG	19	1.79
2021	NYG	162	3.66
2022	NYG	295	4.45
2023	NYG	248	3.89
2024	PHI	348	5.78
2018 to 2024	Career	1,552	4.66
2025, weeks 1 to 5	Eagles	83	3.22

nflverse play-by-play, regular season 2018 to 2025, designed runs with a named rusher

? FIVE GAMES INTO 2025 HE IS AVERAGING 3.22 YARDS A CARRY AGAINST A CAREER 4.66. SLUMP, OR CLIFF?

HOW OFTEN DOES AN ESTABLISHED BACK FALL OFF A CLIFF?

SEASON	ESTABLISHED BACKS	FELL A FULL YARD
2020	8	0
2021	7	0
2022	11	0
2023	11	2
2024	11	0
2020 to 2024	48	2, or 4.2%

That is the prior: about 4 in every 100 established backs, in any given season.

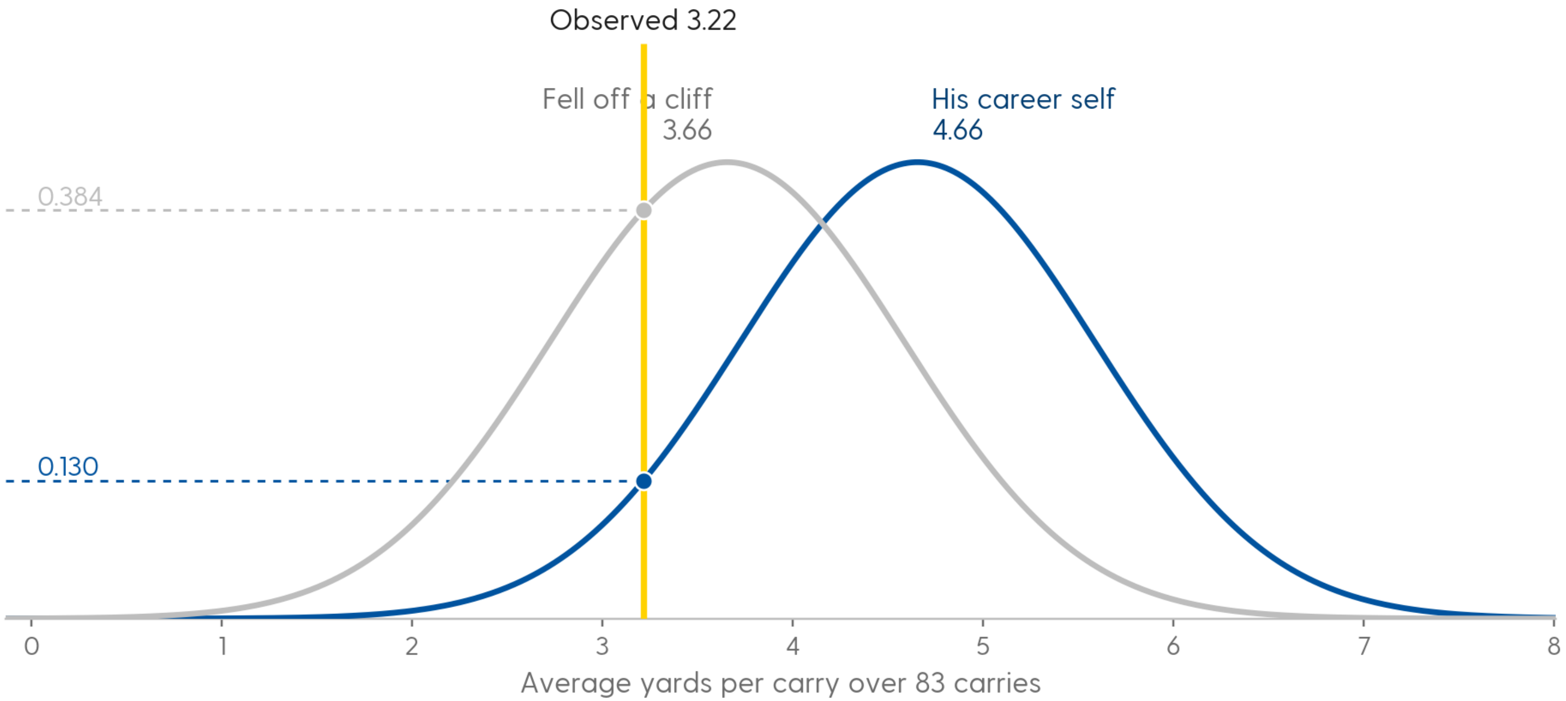
Established: at least 400 carries over the two prior seasons and 100 in the season itself. A cliff: yards per carry at least 1.0 below his career average. nflverse play-by-play, regular season 2018 to 2025, designed runs with a named rusher

PROBABILITY DENSITY

The height of a curve for a continuous variable. It is probability per yard rather than a probability: the chance of an average landing on exactly 3.22 is zero.

Only the area under a stretch of the curve is a probability, which is what a p-value is. Read at the data and compared across hypotheses, the height is the likelihood. Density carries units, so one of them means little alone; a ratio of two cancels the units, which is why every comparison in this deck is one likelihood over another.

HOW LIKELY IS 3.22, UNDER EACH HYPOTHESIS?

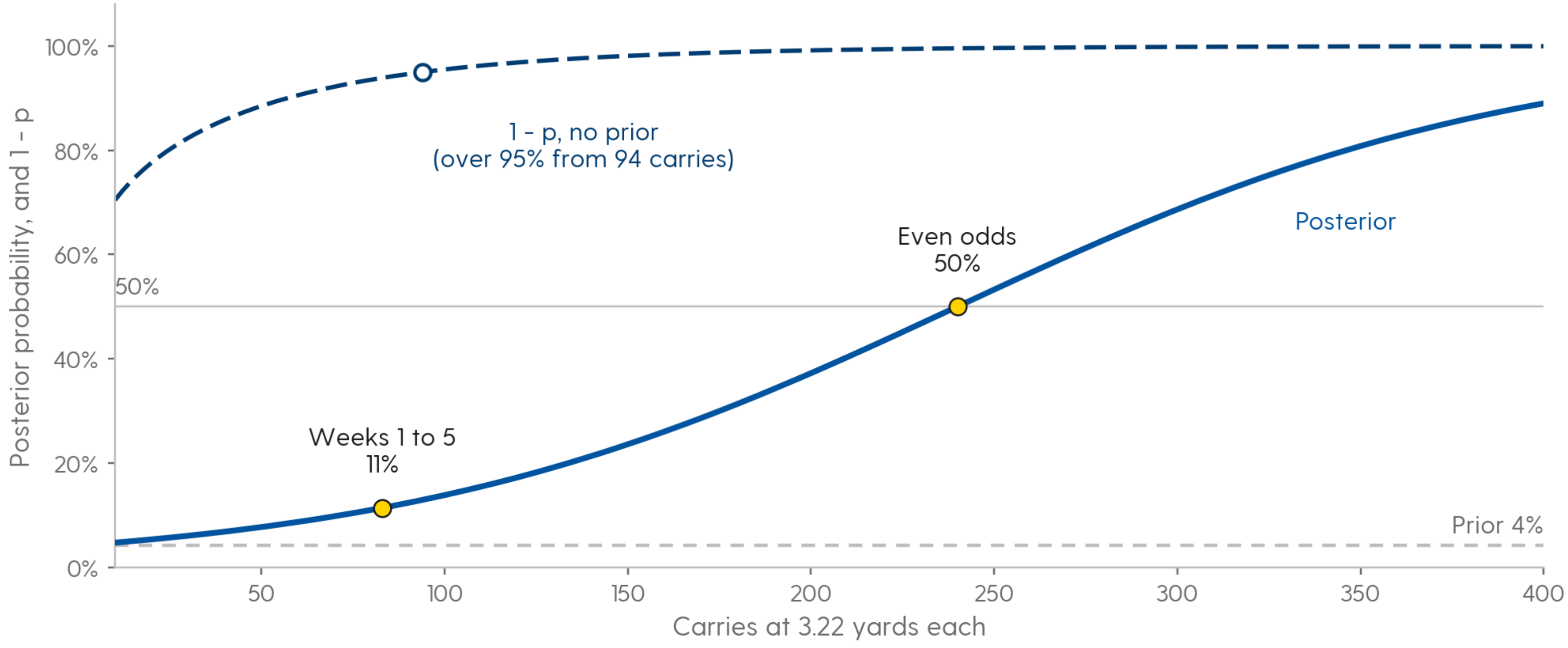


The height of each curve at 3.22 is the likelihood: 0.384 if he fell off a cliff, 0.130 if he is his career self. The data is 3.0 times more likely under the cliff.

Both curves are the sampling distribution of an average over 83 carries, with standard error 0.93. nflverse play-by-play, regular season 2018 to 2025, designed runs with a named rusher

? PRIOR 4.2%. THE DATA IS 3.0 TIMES MORE LIKELY UNDER THE CLIFF. WHAT IS THE POSTERIOR?

HOW MANY CARRIES AT 3.22 BEFORE YOU WOULD BELIEVE IT?



It would take about 240 carries at 3.22 YPC until we would calculate Saquon falling off a cliff to be more likely than not.

Same prior and hypotheses, with the standard error shrinking as the carries add up. The two curves are not the same quantity: 1 - p is a confidence level, and it is not the probability that he fell off a cliff.
nflverse play-by-play, regular season 2018 to 2025, designed runs with a named rusher

NORM.DIST ANSWERS BOTH QUESTIONS

NORM.DIST(x, mean, standard_dev, TRUE) is the area to the left of x.

That is the p-value: the chance of an average this low or lower if he is still himself.

NORM.DIST(x, mean, standard_dev, FALSE) is the height of the curve at x.

That is the likelihood. Compute it once with his career average as the mean and once with the cliff average, same standard error both times.

The posterior is prior x likelihood, divided by the total.

Prior x cliff likelihood, over that plus (1 - prior) x career likelihood.

03

A HOT START

.362

This happens every April. The manager is going to have to decide whether to believe it and adjust his lineup.

Will Smith hit .362 in April 2024, 38 for 105. His career average entering the season was .261 over 1,670 at bats.

MLB Stats API, season and date-range hitting totals. Career is 2010 to 2023.

? IS WILL SMITH A .300 HITTER NOW OR IS THIS JUST WHAT HAPPENS IN APRIL?

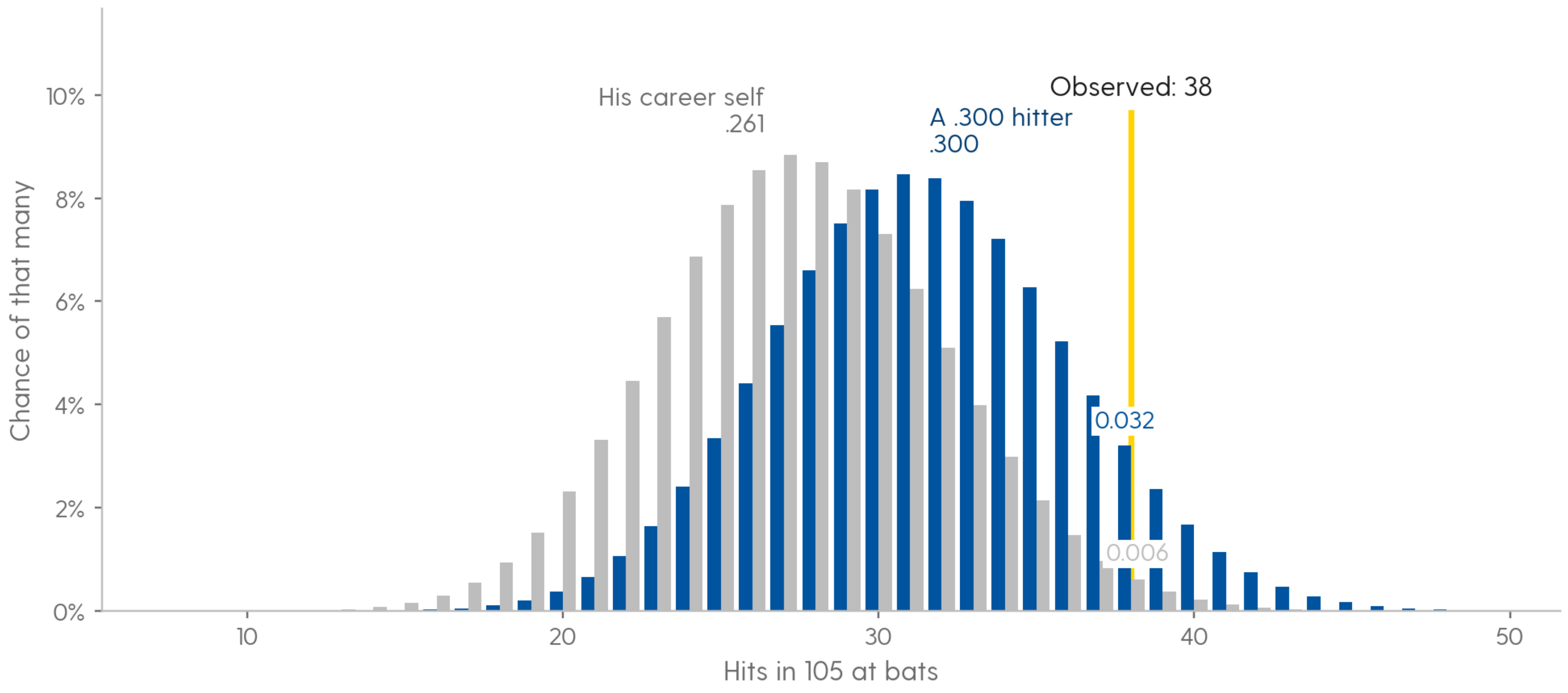
HOW OFTEN DOES AN ORDINARY HITTER BECOME A .300 HITTER?

	2015 TO 2024
Hitter-seasons: career at least 1,000 at bats and .255 to .267 entering the season, at least 300 at bats that season	275
Hit .300 or better that season	14
Share	5.1%

That is the prior: about 5 in every 100.

MLB Stats API, season and date-range hitting totals. Career at bats count from 2010.

38 HITS IN 105 AT BATS, UNDER EACH HYPOTHESIS



0.032 if he is a .300 hitter, 0.006 if he is his career self. His .362 average in April is 5x more likely if he is a .300 hitter than .261.

Two binomial distributions over the same 105 at bats. Hits are a count, so these bars are probabilities rather than densities: the chance of exactly 38 hits is a real number you can quote. MLB Stats API, season and date-range hitting totals

OUR PRIOR WAS 5.1% THAT WILL SMITH WOULD BAT .300 THIS SEASON. THE DATA WAS 5X MORE LIKELY IF HE WAS A .300 HITTER. WHAT IS YOUR POSTERIOR?

MAY

May: 18 for 85, .212

Yesterday's posterior is today's prior: P(.300 hitter) = 0.219

Likelihood of May if .300: 0.0196. If .261: 0.0602

P(.300 hitter | April and May) = 0.084

THROUGH MAY 31 HE WAS HITTING .295. THE POSTERIOR HAD ALREADY GONE BACK TO WHERE IT STARTED.

HOW MUCH OF THIS ANSWER IS THE PRIOR?

PRIOR P(.300 HITTER)	PRIOR ODDS	ODDS MULTIPLIED BY	POSTERIOR ODDS	POSTERIOR AFTER APRIL
1.0%	1:99	5.2	5.2:99	5.0%
5.1% (base rate)	1:18.6	5.2	5.2:18.6	21.9%
10.0%	1:9	5.2	5.2:9	36.8%
25.0%	1:3	5.2	5.2:3	63.6%
50.0%	1:1	5.2	5.2:1	84.0%

Odds = P / (1 - P). Our observed April multiplies our prior odds by 5.2.

? WHAT DID HE HIT THE REST OF 2024?

BINOM.DIST ANSWERS BOTH QUESTIONS

BINOM.DIST(hits, at_bats, p, FALSE) is the chance of exactly that many hits.

That is the likelihood. Compute it once with p = .300 and once with his career average.

1 - BINOM.DIST(hits - 1, at_bats, p, TRUE) is the chance of that many or more.

That is the p-value, with p set to his career average.

The posterior is prior x likelihood, divided by the total, exactly as before.

04

BEYOND SIGNIFICANCE

EVIDENCE SHOULD BE CONSIDERED, BUT OUR PRIOR BELIEF SHOULD IMPACT HOW WE INTERPRET EVIDENCE.

STATISTICAL SIGNIFICANCE

If the null hypothesis were true, how surprising is my data?

Ignores prior beliefs about how unlikely the alternative is.

POSTERIOR PROBABILITY

Given my data, how likely is this hypothesis to be true?

Incorporates prior beliefs about how plausible each hypothesis was before seeing the data.

Coaches ask the second question.

WHERE A PRIOR COMES FROM

A base rate in the data.

2 of 48 established backs fell a full yard. 14 of 275 ordinary hitters became .300 hitters. Count the population the player belongs to.

What the people who watch him already believed.

A scout who saw a swing change in spring training has a prior above the base rate. Document important priors in advance to prevent hindsight bias.

OTHER CONSIDERATIONS

We examined the likelihood a player dropped off a cliff of a full yard or that the hot hitter was truly a .300 hitter. What nuance is this missing?

What other information would you want to look at when determining priors?

05

THIS WEEK'S WORK

SALVADOR PEREZ HIT .355 IN APRIL 2024, 38 FOR 107

1. **Open the hot start sheet in the worksheet.**

   His career average entering the season, .267 over 5,290 at bats, and his April are given. Everything below comes from those cells.

2. **Compute the p-value, then the two likelihoods, with BINOM.DIST.**

   One likelihood for a .300 hitter, one for his career self.

3. **Compute the posterior with the base rate as the prior, then again at 20%.**

   Would you place more weight on the p-value or your posterior when advising the manager?

NICK ANDERSON, AGAIN

Last week you computed how unlikely four straight misses were for a 70.4% shooter: 0.0076. That was the frequentist question.

This week, the question you wanted to ask: how likely is it that he choked? Open the free throws sheet. Assume a player who chokes shoots 50% from the line.

Choose your own prior for a player choking in that spot and put it in B7. Then compute the two likelihoods and the posterior.

Be ready to say what prior you chose, why, and whether the four misses moved you.

CHRISTIAN MCCAFFREY AVERAGED 3.10 YARDS A CARRY THROUGH WEEK 5 OF 2025

1. Open the yards per carry sheet.

   His career, 4.83 over 1,233 carries with a standard deviation of 7.40, and his 91 carries this season are given.

2. Compute the standard error, z, and the p-value with NORM.DIST.

   This is the Week 2 computation.

3. Compute the two likelihoods with NORM.DIST, then the posterior.

   The prior is the base rate from today's deck. Would you say cliff?

YOU WORK FOR THE SEATTLE MARINERS, AND THE MANAGER WANTS TO KNOW WHETHER JORGE POLANCO'S APRIL IS REAL

1. Before a single at bat in 2025, how likely was it that Polanco would hit .300 this season, and why is that the right starting point?
2. After April, how much should the manager's view of Polanco move, and why that much rather than more or less?
3. Of the hitters who started hot, which ones should the manager believe, and what separates them from the rest?
4. Slide the prior in your visual. Where does the answer stop depending on the prior, and what does that mean for how confident you can be?
5. Why is the same April average stronger evidence for one hitter than for another?
6. What else would you want to know before changing the lineup?

The data is weeks/week03/data/april-2025-hot-starts.csv: every established hitter's April 2025 and his career through 2024. Your brief is written to the manager. python3 scripts/new_brief.py week03 writes BRIEF.md with these as headings; /coach-brief 3 critiques a draft and will not write one.

BEFORE YOU PUSH ON MONDAY, SEPTEMBER 14

Commit the filled-in worksheet in weeks/week03/data/.

Commit your analysis code and its outputs in weeks/week03/outputs/.

Answer every question in BRIEF.md, in your own words.

Run /audit, then /quiz-me.

Upload your repository's link to Canvas by Monday, September 14 at 11:59pm.